

# JEE MAIN REVISION MASTER: CAPACITOR

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 167 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                 | Frequency | Category        |
|---------------|------------------------------|-----------|-----------------|
| Very High     | Capacitor - Core Application | 167 PYQs  | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Capacitor - Core Application (Very High Yield — 167 PYQs)

Core Principle: Key problem pattern in Capacitor.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A sphere of capacitance  $100\text{ pF}$  is charged to a potential of  $100\text{ V}$ . Another identical uncharged metal sphere is brought in contact with the ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "A parallel plate air capacitor has a capacitance  $C$ . When it is half filled as shown in figure with a dielectric constant  $K=5$ , the perc..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "A parallel plate capacitor has capacitance  $C$ , when there is vacuum within the parallel plates. A sheet having thickness  $\frac{1}{3}r$ ..." [Ans: D]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)**

A parallel plate air capacitor has a capacitance  $C$ . When it is half filled as shown in figure with a dielectric constant  $K=5$ , the percentage increase in the capacitance is \_\_\_\_\_.

- (A) 33.34
- (B) 66.67 [Correct]
- (C) 200
- (D) 400

**Correct Answer: B**

**Solution Analysis:** For a parallel plate capacitor with plate area  $A$ , separation  $d$ , and air (or vacuum) between the plates, the capacitance is given by :  $C = \frac{\epsilon_0 A}{d}$  For a parallel plate capacitor with plate area  $A$ , separation  $d$ , and dielectric material with dielectric constant  $K$  between the plates, the capacitance is given by :  $C' = \frac{K\epsilon_0 A}{d}$  When a dielectric slab of thickness  $t = (d / 2)$  and dielectric constant  $K=5$  is inserted, the system is with two capacitors connected in series : Capacitor 1 (  $C_1$  ) is filled with dielectric, thickness  $d_1 = (d / 2)$